

VN-Index Sector Analysis: What Drives Returns in Vietnam's Stock Market?

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1. Abstract

This report examines the impact of major sectors on the VN-Index's performance from 2019 to 2025, focusing on Banking, Real Estate, Consumer Staples, and Information Technology. The analysis covers a significant phase in Vietnam's equity market development, including the COVID-19 pandemic, the 2022 corporate bond and real estate crisis, global macroeconomic challenges, and a robust market recovery in 2025.

Banking remains the primary market anchor due to its high correlation with the VN-Index, substantial explanatory power, and sensitivity to market fluctuations. Information Technology, exemplified by FPT, is the most potent growth driver, delivering favorable risk-adjusted returns and surpassing the broader index. Real Estate has the weakest risk-return tradeoff due to its high volatility and structural constraints after the 2022 bond market scandal. Consumer Staples provides defensive characteristics but generates limited return growth.

In summary, the VN-Index's performance reflects the interplay between stable market leadership from Banking and high-growth contributions from Information Technology, while Real Estate and Consumer Staples assume tactical roles. Investors should consider Banking for market exposure, Information Technology for long-term growth allocation, Real Estate for high risk tolerance, and Consumer Staples as a short-term defensive hedge.

2. Introduction

2.1. Motivation and Project Scope

Vietnam is rising as one of Southeast Asia's fastest growing economies, making the equity market an interesting lens on emerging market development (Alimov 2026). This project aims to unpack the VN-index in a structural way in terms of sector drive and risk factors in this 2019–2025 window, capturing a meaningful maturation phase with significant changes and improvements. This paper examines which sectors primarily drive HOSE returns and whether this shifts across fast-moving market conditions, alongside the risk-return characteristics of each sector to draw implications for portfolio construction.

VN-Index is the measurement of Vietnam equity market, including 3 major indexes: HOSE, HNX and UpCom. This paper focuses on HOSE as the leading index by market capitalisation and liquidity, covering large-sized entities. As of September 2025, HOSE listed 696 stocks with a cap value exceeding VND 7.20 million billion, equivalent to 62.58% of GDP (SSC 2025). Despite this, Vietnam ranked last among SEA peers in market cap as % of GDP as of 2022, highlighting significant headroom for growth (TheGlobalEconomy.com 2024). The market is retail-heavy, with individual investors accounting for over 90% of daily trading volume, which injects liquidity but also increases volatility due to their short-term, speculative trading behavior.

HOSE is a market-cap weighted index, where larger companies exert disproportionate influence on index movements. The index faces several accessibility constraints, notably foreign ownership limits and low free-float ratios driven by the dominance of state-owned enterprises (SOEs) (Collins 2026). Several mechanisms have been introduced such as the new Resolution 79, alongside Law No. 56/2024/QH15 to address these issues.

2.2. Data description

Four sectors Banking, Real Estate, Consumer Staples and Information Technology (Info Tech) are chosen because they're dominant player in the industry in terms of market caps, consistent with MSCI's Vietnam classification (MSCI 2026). Banking and Real Estate are the two most mature and dominant sectors in Vietnam economy and stock market, but recently undergo some reforms and regulatory adjustments relating to credits and foreign investment towards more sustainable growth. Consumer Staples provides a more stable factor into understanding non-cyclical behaviors of stock market, while Info Tech is a growing and constantly changing sector, especially in a developing economy like Vietnam. This choice of sectors provides a balanced view of market cap, liquidity, momentum and volatility.

Vietcombank (VCB) and Sacombank (STB) are selected to be proxies for Banking sector. While VCB is inarguably the most large-cap, high-quality segment of the financial sector, STB reflects strong retail and speculative interest with high momentum. VCB as a state-owned enterprises compared to STB in private sector also provides more depth in constructing the banking index.

For Real Estate, Vinhomes (VHM) and No Va Land (NVL) are chosen, despite VIC being a higher-market-cap stock in this sector. Since VHM is a Vingroup subsidiary, including VIC would introduce double-counting and skew performance through idiosyncratic conglomerate risk. NVL is selected instead for its decent market cap and focused exposure to residential real estate with domestic institutional backing.

Vinamilk (VNM) and Masan Group (MSN) are chosen for Consumer Staples sector. While VNM is more of a stable stock with dominant market cap in the dairy industry, MSN captures the growth and momentum with a broad structure as a conglomerate, ideal for a broader consumption proxy. Together, they capture the two essential pillars of Vietnamese consumption: dairy/nutrition and packaged goods/retail.

For Info Tech sector, FPT is used as a single-stock proxy given the thin technology representation on HOSE, which itself reflects the nascent stage of Vietnam's listed tech sector.

This project uses daily closing price in local currency VND from Jan 2019 to Dec 2025 from [Investing.com](https://www.investing.com). Considering the data selection, there may be unsystematic risks distorting the sector data since sectors are constructed using top representatives. Selection bias is also visible since we ignore low-market-cap and small firms in the sectors.

2.3. Variable Construction

Log returns are used throughout the analysis as they are stationary by construction, satisfying OLS assumptions without requiring differencing. Sector returns are constructed using a value-

weighted approach based on adjusted market capitalisation (see Appendix A). Data cleaning are also applied in terms of formatting price, volume and date figures.

3. VN-Index Analysis

3.1. VN-Index Price and Return series

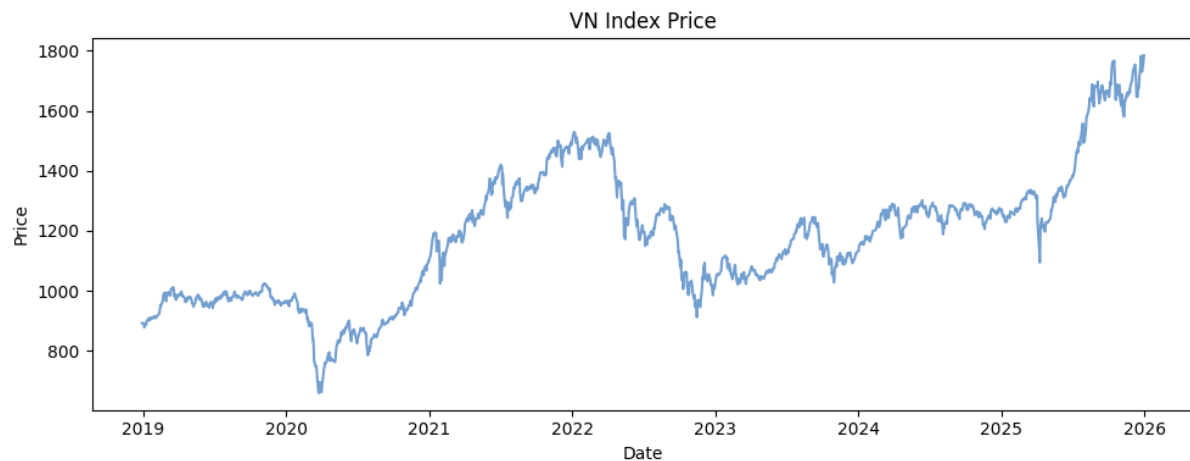


Figure 1: VN-Index Price Series

VN-index showed an overall upward trend. Sudden shock occurred in Q1 2020 due to the outbreak of COVID-19, bottom out at 650-700. However, during the lockdown period, the stock market observed a swift recovery driven by pandemic policy confidence, underpriced valuations, and a retail investor boom through late 2021 (Hung, Hue & Duong 2021). In 2022, VN-Index fell from approximately 1,500 in April to below 900 by November, a drawdown of roughly 40%, pressured by the geopolitical and global economic effects like Ukraine-Russia war and US inflation, following by a series of major scandal related to bond market and stock manipulation by Van Thinh Phat, Tan Hoang Minh and FLC.

By 2025, the index broke above its 2022 peak and reached all-time highs around 1,750-1,800. 2025 illustrates a major upward trend amid uncertainties in the global economy thanks to Vietnam's stable economic growth and an ideal policy environment, marking the pivotal year of post-shock recovery.

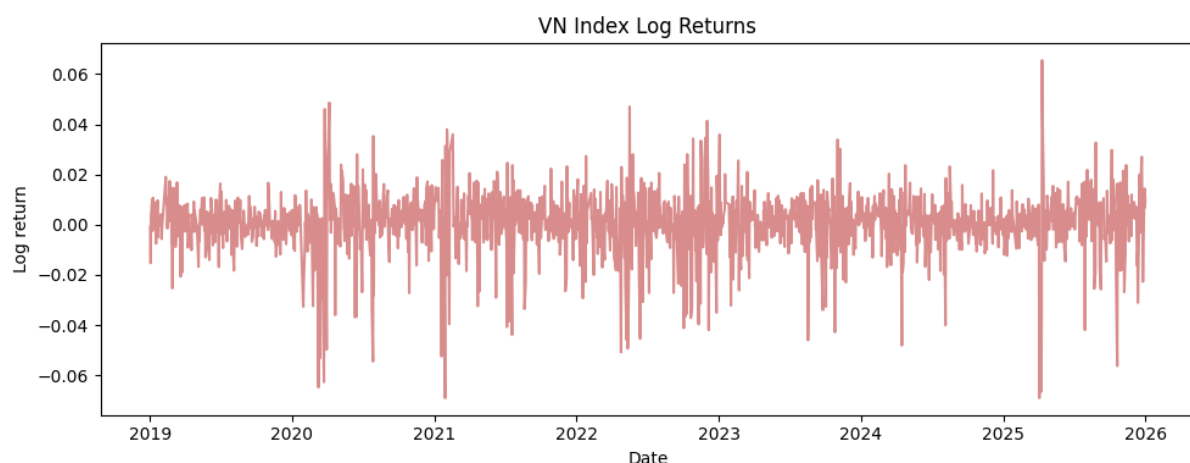


Figure 2: VN-Index Log Return Series

Regarding the return series, VN-index show a dense and frequent movements, reflecting high trading activity and noise. Pre-COVID volatility was relatively contained at $\pm 0.02\%$, but shifted to a higher regime post-2020, with clustering around $\pm 0.04\%$ and large moves grouping together during stress periods. A notable spike in Q2 2025 stemmed from fears over Trump's tariff hikes in Vietnam, though the index rebounded within the week. Negative returns are more prominent than positive ones, consistent with negative skewness and asymmetric downside risk.

3.2. Summary Statistics

VN-index daily returns exhibits a mean of 0.0396% and standard deviation of 1.23%. Annually, this represents a mean of 10.5% and volatility of 19.5%, illustrating substantial volatility and non-normality. The negative skewness of -1.07 confirms the visual observation that large negative return spikes appear more extreme than positive ones, while kurtosis of 5.58 confirms volatility clustering with returns exhibiting fat tails inconsistent with normality. Considering these characteristics, VN-Index is consistent with emerging market equity returns.

4. Sector Performance

4.1. Returns and Volatility

	Mean Return	Std Dev	Skewness	Kurtosis	Annualised Return	Annualised Volatility	Sharpe Ratio
Banking	0.05%	1.71%	-1.95	23.52	12.39%	27.15%	0.46
Consumer Staples	-0.01%	1.59%	-0.50	4.33	-2.82%	25.17%	-0.11
Real Estate	0.01%	1.79%	-0.09	2.39	3.46%	28.36%	0.12
Info Tech	0.11%	1.72%	-0.12	3.27	28.03%	27.34%	1.02

Figure 3: Summary Statistics for Each Sector

Among the four sectors, Info Tech delivers the best overall risk-return performance with the highest annualised return of 28.03% at comparable volatility to peers. Banking also performs relatively well at 12.39% annualised return but carries the heaviest tail risk, reflected in deeply negative skewness of -1.95 and kurtosis of 23.52.

Real Estate performs weakly with the worst risk-return tradeoff: highest volatility at 28.36% yet near-zero returns of 3.46%. This suggests that Real Estate experiences relatively high uncertainty without generating equally strong returns, making it the least efficient sector. Consumer Staples is the weakest-performing sector over the sample period with negative annualised returns of -2.82% despite the lowest annualised volatility of 25.17%.

In terms of risk-adjusted performance, Info Tech and Banking both generate compensated risk, delivering positive returns relative to the volatility taken. Info Tech leads with the highest Sharpe Ratio of 1.02, meaning investors were rewarded most effectively for bearing risk. Banking also offers compensated risk yet less efficiently than Info Tech. Real Estate provides weak compensated risk as the Sharpe Ratio is only 0.12. In contrast, Consumer Staples reflects uncompensated risk with negative Sharpe ratio of -0.11.

The correlation matrix shows that all sectors are positively correlated, moving in the same direction as broader market conditions, though only at moderate levels, suggesting sector diversification still provides some benefit. Banking shows the strongest co-movement with Consumer Staples (0.495) and Info Tech (0.487). Real Estate is the most distinct sector in the sample, with the lowest correlation against Info Tech at 0.365, meaning a portfolio combining these two offers the greatest diversification benefit.

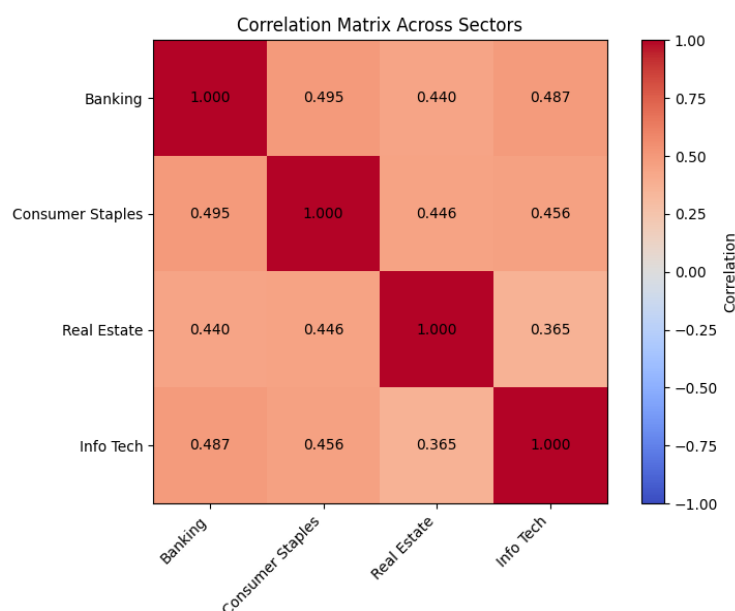


Figure 4: Correlation Matrix across Four Sectors

4.2. Cumulative Return Analysis

Info Tech is the dominant growth contributor to VN-Index cumulative returns, driven by the high-growth, high-liquidity nature of the sector. FPT as the representative of the index exemplifies this through its software export operations and leading role in Vietnam's digital transformation, delivering a cumulative return of approximately 2.0 by the end of 2025, roughly 3x the VN-Index return (DNSE 2025a).

Banking displays a fairly stable performance, tracking closely above the VN-Index, which is consistent with its high index weight mechanically linking its returns to the broader market. Banking's outperformance was most pronounced during the 2023-2024 stress period. As the market normalised in 2025 and other sectors recovered, its relative advantage diminished.

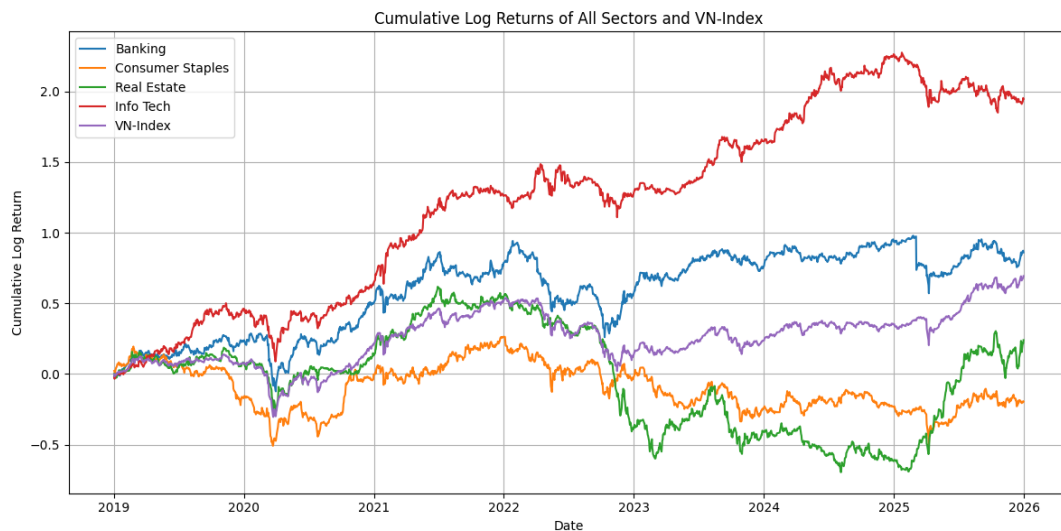


Figure 5: Cumulative Log Returns of VN-Index and Sectors

Real Estate tracked the VN-Index until the bond market scandal in Q2 2022, which triggered a collapse in returns in Q2 2022. The crisis deepened in the following quarters with the arrests of Van Thinh Phat and Tan Hoang Minh on charges of illegal bond issuance and securities fraud, exposing systemic vulnerabilities in Vietnam's corporate bond market. The simultaneous loss of investor confidence in corporate bonds and real estate equity triggered a dual shock, ushing cumulative returns negative. Despite partial recovery, Real Estate remains the only sector that destroyed value over this period. No Va Land was heavily implicated in the 2022 bond market turmoil, which distorts the real estate sector proxy during this period. However, it also more accurately represents the funding vulnerabilities of mid-tier developers compared to Vinhomes, which benefits from Vingroup's stronger balance sheet and brand dominance.

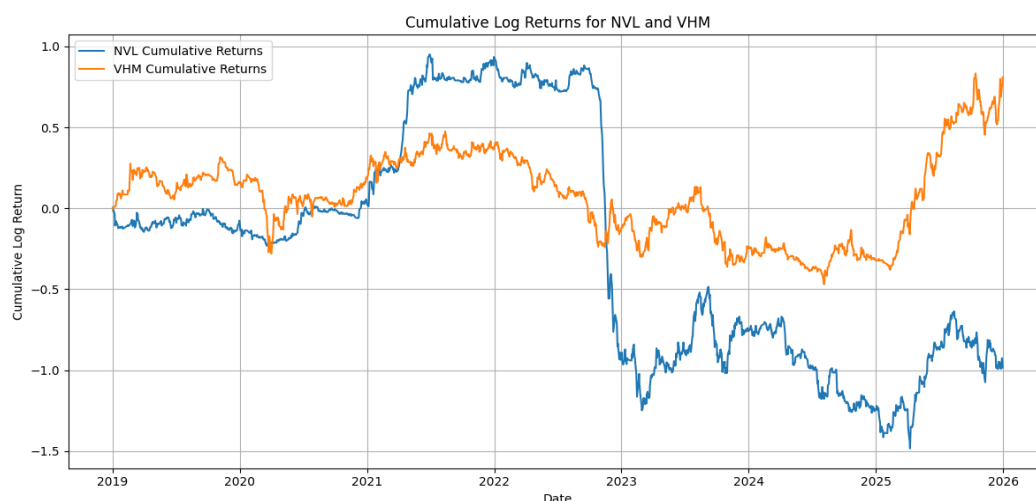


Figure 6: Cumulative Log Returns of Real Estate breakdown

Consumer Staples is the most stable but persistent underperformer, with cumulative returns fluctuating around zero, consistent with its defensive nature and limited sensitivity to most shocks outside of COVID-19. Structural headwinds such as market saturation and an ageing population reducing daily essential spending resulted in minimal growth in this industry. The slight recovery in 2025 illustrated the potential of product diversification and improvement in

profit margins from F&B companies, especially with the leading companies like Masan or Vinamilk (DNSE 2025b).

Therefore, VN-Index's modest cumulative return reflects the averaging effect of divergent sector outcomes. Starting the period clustered together under COVID-19 pressure, sectors later diverged significantly. The 2022 scandal dragged most sectors down, with Real Estate most severely affected, while Banking showed relative resilience and Info Tech experienced a temporary stagnation. Overall, Info Tech and Banking are the primary drivers of positive cumulative returns for VN-Index across the study period.

5. Correlation Dynamics

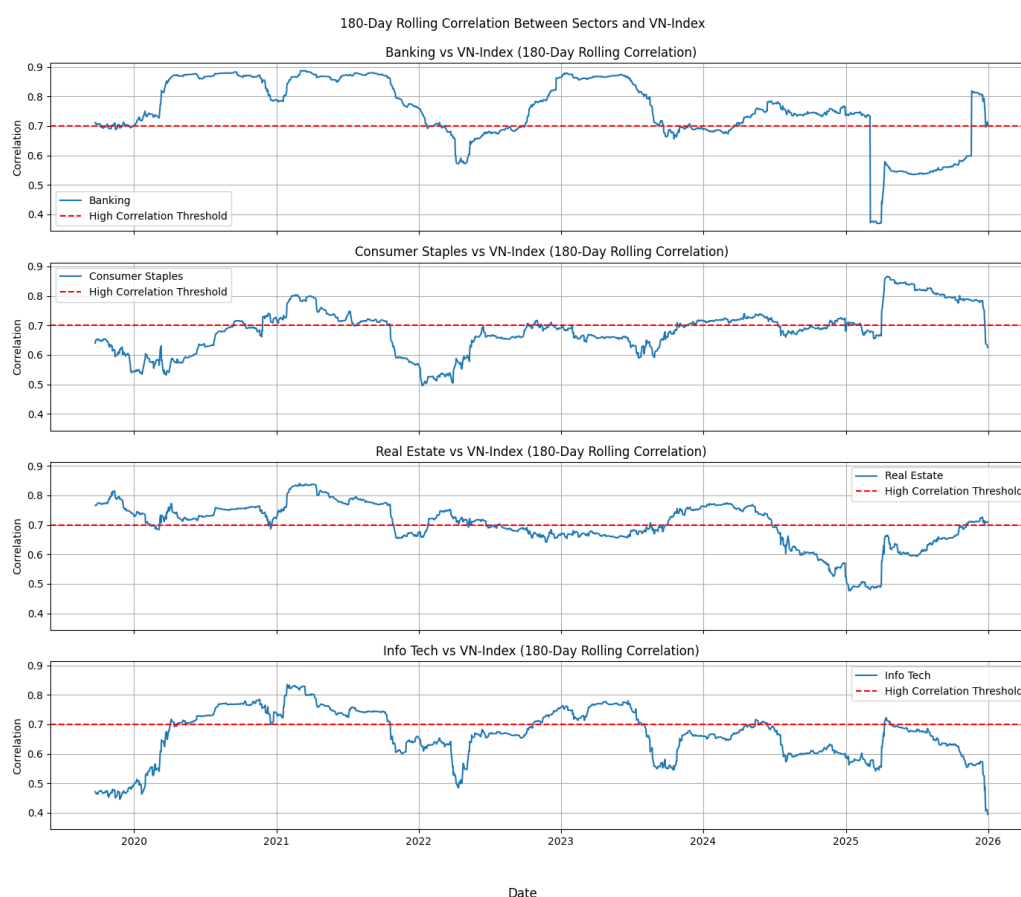


Figure 7: 180-day Rolling Correlation Between Sectors and VN-Index

Overall, relationships between sectors and VN-Index fluctuate significantly throughout the period but converge during stress events such as COVID-19 and the 2022 scandals. Banking is the most informationally connected to the broader market, diverging briefly in early 2025 before a swift recovery. Real Estate correlates consistently with the market, diverging only in mid-2024 before returning to strong co-movement by end-2025. Consumer Staples and Info Tech both show moderate correlation, with Info Tech declining to an all-time low of ~40% in 2025 due to its self-driven international growth.

Interestingly, in Q1 of 2025, the market observed a bullish pattern as FTSE upgrade news drove broader market gains, which resulted in elevated correlations in Consumer Staples, Real Estate and Info Tech. However, Banking diverged in the opposite direction due to foreign

investor capital withdrawal driven by unfavourable global macro conditions, exchange rate pressure, and limited improvement in market accessibility (Thu Minh 2026). Nevertheless, Banking's swift recovery in correlation by end-2025 reinforces its role as the primary market-connected sector throughout the study period.

6. Regression Analysis

This regression analysis examines how sector log returns respond to movements in the VN-Index, with beta measuring market sensitivity, alpha capturing unexplained returns, and R^2 indicating how much of sector variation is driven by systematic versus idiosyncratic factors.

Sector	Alpha	Beta (R_market)	Beta (R_market_lag1)	Beta (R_sector_lag1)	R^2	Adj R^2	Alpha p- value	Market p- value	Market Lag p- value	Sector Lag p- value
Banking	0.000098	1.0528	-0.0591	0.0116	0.5665	0.5658	0.7161	0.0000	0.0777	0.6275
Consumer Staples	-0.000456	0.8862	-0.0387	0.0453	0.4701	0.4692	0.0993	0.0000	0.2119	0.0581
Real Estate	-0.000233	1.0063	-0.1240	0.1492	0.4903	0.4894	0.4454	0.0000	0.0003	0.0000
Info Tech	0.000773	0.9296	-0.1393	0.0354	0.4381	0.4371	0.0127	0.0000	0.0000	0.1378

Figure 8: Regression Results of Sectors with VN-Index

Banking is the most market-sensitive sector, with the highest beta meaning it tends to amplify market movements in both directions. Its high R^2 suggests that most of its risk is systematic, with no significant alpha indicating little evidence of abnormal return. Real Estate behaves similarly to the broader market, with a beta close to one and moderate explanatory power though lacking significant alpha and reacting slower to market information. Consumer Staples is the most defensive sector, with a below-one beta and insignificant alpha, meaning its returns are largely explained by market exposure with little independent performance. Info Tech stands out as the only sector with a significant positive alpha, suggesting it generates returns beyond what market movements alone can explain. Its low R^2 confirms that a larger portion of its returns is driven by sector-specific factors rather than systematic risk, offering meaningful diversification benefits alongside its below-one beta.

The scatter plots also support the regression results. Banking and Real Estate show steeper fitted lines, which matches their higher beta values. Banking also has points that are more closely clustered around the fitted line, supporting its higher explanatory power. In contrast, Info Tech shows a wider spread of points around the line, which is consistent with its lower R^2 and higher idiosyncratic risk.

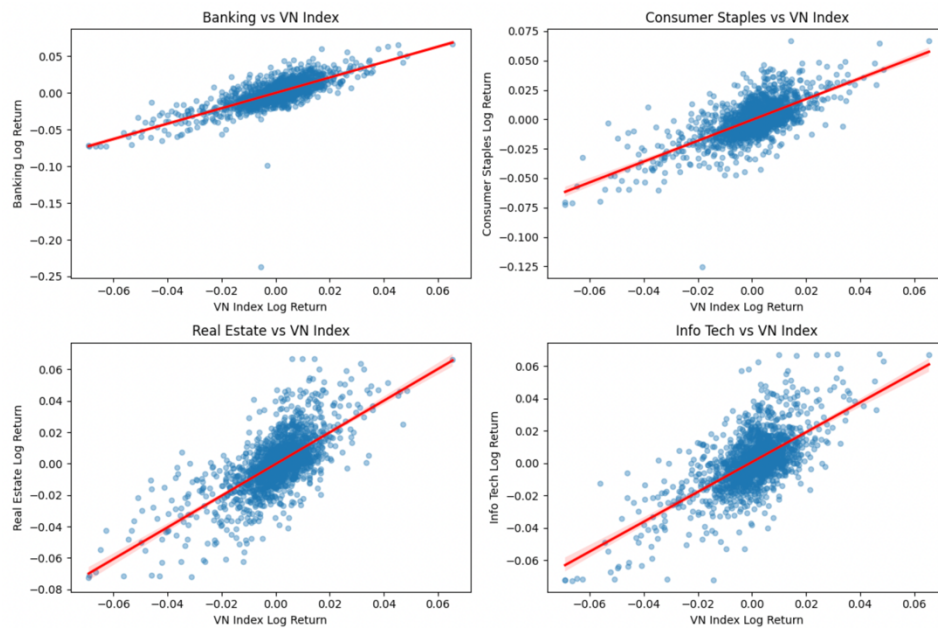


Figure 9: Scatterplot of Sectors with VN-Index

These findings carry clear implications for portfolio construction. Banking suits investors seeking stronger market exposure, though it carries higher systematic risk. Investors should therefore be aware of the drawdown risk of Banking during market downturns. Real Estate offers limited downside protection during downturns given its close co-movement with the market. However, its delayed market reaction suggests a potential short-term trading opportunity, where investors could anticipate Real Estate movements based on prior market signals. Consumer Staples appears more defensive, helping reduce portfolio risk and act as a reliable hedge during market stress. Info Tech, with its below-one beta and high idiosyncratic returns, offers genuine diversification benefits. Overall, combining high-beta sectors such as Banking with more defensive sectors such as Consumer Staples and Info Tech may help create a more balanced portfolio. The spread in R^2 across sectors also allows investors to explicitly allocate between systematic and idiosyncratic risk when budgeting for portfolio risk exposure.

7. Volatility Analysis

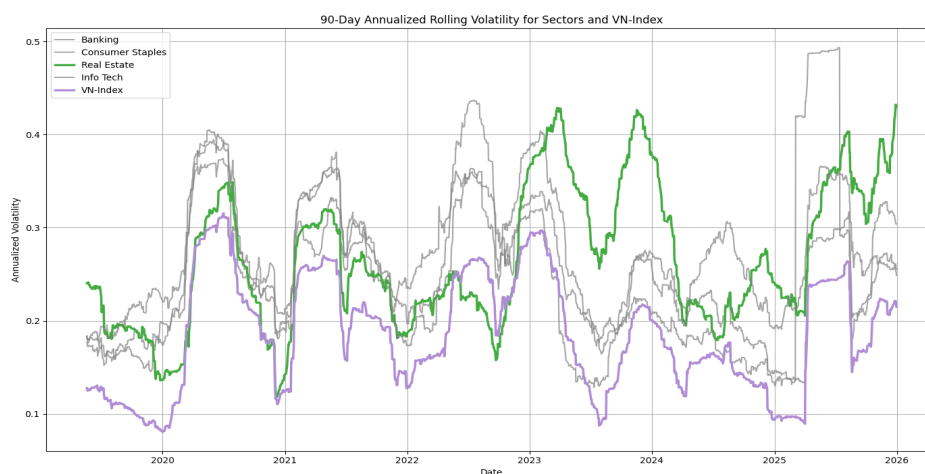


Figure 10: 90-day Rolling Volatility for Sectors and VN-Index

Volatility clustering is clearly evident, concentrated during crisis periods including the COVID-19 outbreak in 2020-2021, the 2022 bond scandal, and the early 2025 market rally where rapid price appreciation generated elevated short-term volatility. Real Estate stood out as the most persistently volatile sector, remaining structurally elevated through 2023-2024 even as other sectors returned to calmer regimes, with a further uptick in late 2025. The VN-Index consistently sits below all individual sector volatility levels, confirming the diversification effect of holding a broad market portfolio over any single sector.

8. Discussion

8.1. VN-index Drivers

Banking is fundamentally the market anchor due to its dominance in the capital market structure and strongest correlation. It's also the most market-sensitive sector with high level of systematic risk and strong explanatory power towards VN-index. Banking's overall performance is also fairly stable, which helps stabilise the VN-index cumulative returns during stress periods. On the other hand, Info Tech is rising as the main growth driver for VN-index with its compounding return well above the market across this period. Nonetheless, it tends to decouple from the market short-term movements, driven by its moderate-to-low correlation with VN-index.

This period proved analytically meaningful with unpredictable macro-economic conditions affecting the market significantly. Banking remains as the resilient leader of the market across all major shocks such as COVID-19 pandemic and the 2022 bond scandal. Info Tech exhibited exceptional growth despite domestic shocks. Real Estate transformed from a market-correlated sector into a structurally isolated industry since 2022, fundamentally shaped the index's return until the end of the period.

8.2. Investment Implications

Info Tech is the most attractive sector overall, offering the best risk-adjusted returns and diversification benefits through its sector-specific growth drivers. However, investors should still note its greater sensitivity to global tech conditions than domestic ones. Overall, it best suits growth-oriented, long-term investors as the growth allocation of a portfolio. Banking suits macro-oriented traders given its close tracking of market movements and predominantly systematic risk profile, making macro conditions more relevant than stock-specific factors. It serves well as a base portfolio allocation, providing market exposure with slightly higher returns than the index. Despite heavy tail risk during stress periods, its risk remains compensated, though careful monitoring is still essential. Given Info Tech's tendency to detach from broader market movements recently, Banking appears to be the most optimal pair with Info Tech.

Real Estate is unsuitable as a core holding given its uncompensated volatility, regulatory sensitivity, and capital destruction since 2022. For speculative investors with high risk tolerance and a long time horizon, it may offer an attractive entry point as the sector could flourish post-scandal. Short-term traders may also exploit its lagged reaction to broader market signals. Consumer Staples is only suitable as a short-term defensive hedge against anticipated stress periods, supported by its below-one beta and resilient behavior during the

2022 downturn. Its negative returns and uncompensated risk make it unattractive as any form of holding position, with structural headwinds suggesting limited near-term improvement.

8.3. Limitations

This project exhibits a few inherent limitations. First, due to a lack of data availability, data are drawn from a relatively short and event-heavy sample period, with inconsistency in proxy market capitalisation percentage construction affecting the precision of results. Despite structural changes following major shocks such as COVID-19 and the 2022 bond scandal, sub-periods are treated equally, which may lead to imprecise results. The proxy-based method is also not fully objective, as it ignores medium-to-small enterprises within each sector. Since sectors are constructed using only top representatives, unsystematic risks may distort the sector data. OLS regression is applied, which carries standard limitations such as non-normality of residuals and autocorrelation. Finally, as this project is based solely on historical returns, the forward-looking view of sectors and portfolio implications may not accurately reflect the actual future of each sector.

9. Conclusion

This project investigated the interplay of four core sectors - Banking, Real Estate, Consumer Staples, and Info Tech - in influencing VN-Index movements by employing OLS regression on each sector's returns. The analysis assesses each sector's impact on market movements and its implications for portfolio construction.

Banking and Info Tech emerge as the core holding positions with the greatest market impact, whereas Real Estate and Consumer Staples are better suited for short-term or tactical use. However, the project still carries several constraints regarding data availability, model assumptions, and proxy-based methodology. Future research could address these by utilising machine learning techniques to capture non-linear relationships and complex sector interactions beyond OLS assumptions. A more rigorous methodology for constructing sector returns over a longer period would also provide a more nuanced understanding of sector behaviour and its correlation with the broader Vietnamese capital market, particularly as Vietnam develops toward Emerging Market status.

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Appendix

A. Variable Construction – Sector Weights

Sector	Stock	Weight (%)
Banking	Vietcombank (VCB)	57.86%
	Sacombank (STB)	42.14%
Real Estate	Vinhomes (VHM)	68.25%
	No Va Land (NVL)	31.75%
Consumer Staples	Vinamilk (VNM)	44.80%
	Masan Group (MSN)	55.20%
Info Tech	FPT Group (FPT)	100%